Edge reporting + algorithmic inference

Making unseen shared bikes visible to dispatch

Independent research: how a few off-peak trucks could pre-position bikes, from public data and edge reporting

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Bike-share "no dock to return to": edge reporting and a dispatch algorithm

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The author has a separate Taiwanese patent application (different subject from this method)

Dispatch-level algorithm and visibility only: no buying bikes, no building docks, no operator business decisions

The perennial question, asked differently

The author's observation: station data is a common topic in hackathons and data contests (not counted year by year)

How it is usually asked

Station data is public, so the usual approach is:

Predict which station will run short or full.

All of these assume the same thing:

What the official data can see is all the bikes there are.

This study asks it this way

Station data records only docked bikes.

Bikes parked outside the docks, waiting for someone, are not in the data.

So the question is not only "which station will run short",

but "how should the layer we cannot see be pre-positioned?"

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One public dataset, a different answer

No operator system changes: with public station data and a self-estimated off-dock count, estimate how few off-peak trucks can pre-position bikes along forward routes;

edge reporting (temporary return) only fills in the off-dock layer; it can be computed without it, and with it the estimate becomes a report.

Proof of concept and replay estimates, not operating results; no affiliation with any operator.

Stations grow faster than bikes, so the shortage persists

Two pieces of evidence together: ① official figures show stations growing faster than bikes; ② after the NT$5 ride vouchers the counts fell, but the shortage remains

Stations and docks grow fast while bikes grow slowly: new stations can only draw on the existing pool of bikes, and the data show a long-standing shortage, not a one-off event.

Evidence ①: official and the author's figures	Stations / docks	Bikes
End-2023 → end-2025 (DGBAS statistical bulletin)	Stations +8.4%	+3.2% (−0.8 per station)
End-2025 → 2026-09-30 (official figure)	Stations +7.3%	+0.0% (bike column may not be updated)
Author's daily snapshots, 07-21 → 10-16	Docks +2.7% (46,180 → 47,446)	Bikes not published officially

Evidence ②: NT$5 ride voucher scheme	Taoyuan transport bureau statistics
"Cannot borrow" per month	15,484 → 11,450 (reported: about 27% better)
"Cannot return" per month	690 → 454 (reported: about 34% better)
Taipei–New Taipei–Taoyuan total (to 2026-05)	4.06 million vouchers issued, 445,000 participants

How to read the two pieces of evidence

Official sources do not publish total docks, so the station count is a proxy; the author's daily snapshots are the only accumulated dock source. Official figures show stations growing faster than bikes; the snapshots show docks growing (bikes are not published).

After the NT$5 vouchers the counts fell, but Taoyuan still has about 11,500 "cannot borrow" and 454 "cannot return" per month: incentives move only part of the bikes, so pre-positioning a few trucks off-peak is worth testing.

Sources: DGBAS bulletin No. 19 (ROC 115); United Daily News 2026-06-24 (Taoyuan transport bureau); author's snapshots. Incentive figures: Taoyuan only.

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Approach: public data, an openly described dispatch algorithm

Independent proposal, not a party to any operator or government contract: use public data to make the unseen visible, then estimate how to dispatch.

Existing data only

Official live station data (bikes, free docks) + the author's own daily snapshots

No operator system changes and no new hardware needed to compute

The edge device only fills in the off-dock layer, and the method works without it

Algorithmic inference

Each station's own usual level is its target

Unit = 2 km shared-reservoir circle; forward routes, few trucks, pre-positioned off-peak

Parameters come from public sources (14 bikes per truck is the operator's figure); off-peak pre-positioning is this study's design; shortage is shown as scenarios

Design constraints

Public data only; edge report fields contain no rider identity

No unlocking, no lock control, no billing changes: observe and report only

Output is algorithmic estimation, not an operator decision; data licensing and privacy are still being verified

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Principle: evidence-based inference only (estimates, not operating results).

How this proposal relates to existing steps: signal source on the bike, physical lock and billing, the full-station process, broken-bike reports; each follows the operator's official process.

The problem: no dock to return to

From the operator's customer–service description and user experience: when a station has no free dock, the bike is left beside it and staff are called; billing is extended meanwhile.

No free dock

No dock to lock into at return time

→

Bike left outside the docks

Locked or pinned, but the return is not registered

→

Invisible to official data

Status stays "rented"; billing extended until staff help

→

Station data disagrees with the field

Data shows a shortage while bikes are actually there

Field example (2026–09–17)

A station in Taoyuan: 25 docks, with another 20 bikes outside (front wheels not in the rail; photographed). At 17:15 the feed showed 2 available while about 22 were on site; on 2026–09–22 about 20 were still outside

Official data records only docked bikes, so these 20 do not exist in the data

The solution in two layers

Make it visible: over Bluetooth, a rider at a full station makes a temporary return at a virtual dock beside it (press the button on the bike unit, pair), so off–dock bikes count toward the level

Make it accurate: predict which stations fill up and when; take bikes out off–peak and free space before the peak (few trucks, forward routes inside 2 km circles)

No unlocking, no lock control, no billing changes: this proposal offers visibility and dispatch suggestions; whether return and billing rules change is up to the operator.

The field example is one station, one moment, counted by eye; it is neither a distribution nor a citywide average. It shows that the phenomenon exists, can be counted, and cannot be expressed by the official station fields.

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Official reporting: returns, full stations and broken bikes

Compiled from operator notices, customer-service statements and user experience; if any step is incomplete, the bike can be in the field while invisible in the data. The operator's notices are authoritative.

① Lock into the dock

Physical lock: front wheel into the rail, locked

② The bike sends a signal

YouBike 2.0: the signal source is the bike itself

③ Payment settles

Fees settle and only then does the status become "returned"

Failed return

Signal not sent or received: bike locked, but the back end never gets the return

Payment fails: status stays "rented" (user experience)

Safety pin in but return not registered: billing extends until staff finish it

Full station (no free dock)

▶ Added here by this proposal: temporary return (Bluetooth, virtual dock)

Operator steps: lock beside a free dock → "no space" in the app → apply for time extension → phone in; staff come

Billing extends until staff complete the return

Broken-bike report

App "report repair" (Taipei)

Seat reversed 180° so patrol staff can see; a faulty bike returned to the same station within 5 minutes is free

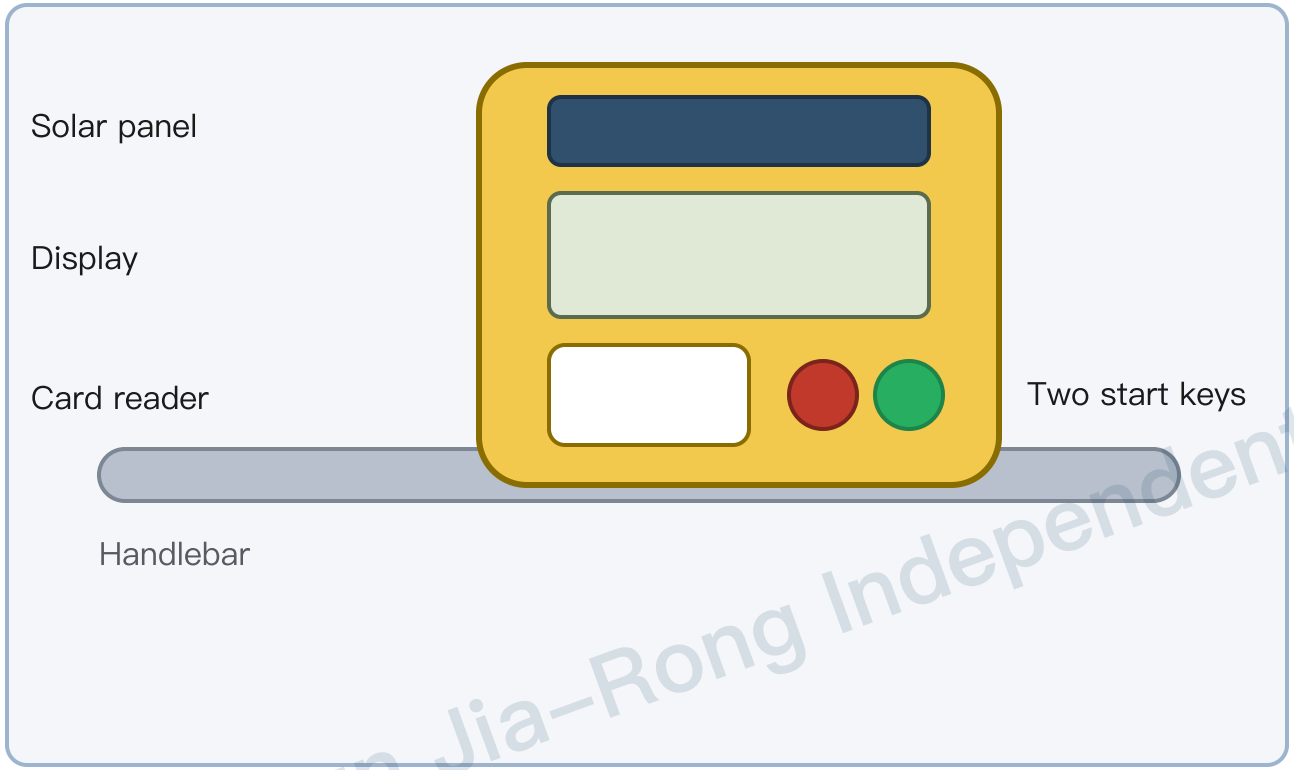
What this proposal could add: temporary return (a concept, no operator agreement)

A rider at a full station registers a temporary return at a virtual dock over Bluetooth, recorded in this study's own log so off-dock bikes count toward the level. Locks and billing stay with the operator.

Sources: TVBS 2024-08-14 (full-station process; Taipei's current statement is a 15-minute "full-station extension"); Taipei transport bureau FAQ (broken-bike reports). Details may have changed.

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The bike unit: the bike itself is the signal source



What the illustration shows

The unit sits at the handlebar centre: display, card reader, two start keys, and a solar panel on top

A label says the bike may only be returned to its dedicated dock

According to the Taipei transport bureau, the 2.0 main unit is on the bike and the 1.0 unit was in the dock: the signal source is the bike itself

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Illustration: drawn by the author from public photos and reports; not an operator drawing and uses no third-party photo or logo.

What this means for the proposal

The concept unit would likewise sit on the bike with the bike as signal source, plus one button to pair with the virtual dock; bench only, not to be installed on operator bikes without authorization.

Possible origins of off-dock bikes: related cases

Case	What happened	Source	What this means for the proposal
Full station, no space (2024-08)	Customer service advises locking the bike, choosing "no space" in the app to apply for a full-station extension and phoning it in; billing extends until staff help complete the return	TVBS 2024-08-14	Return registration depends on manual fixes: the bike is in the field while the data has a gap
System outage (2025-10-01)	Evening peak in Taipei, New Taipei and Taoyuan: more than 20,000 users stranded, neither regular nor electric bikes usable; the operator cited a 9/27 system update causing a traffic surge and back-end connection failures; restored before 19:50 per the Taipei transport bureau	TVBS English 2025-10-02	When the back end fails, return and rental events may not be recorded
Nationwide outage (2024-06-11)	Nationwide system outage; at the time of reporting the operator said the cause was being confirmed, later linked to an overnight update	FTV English News 2024-06	Public reports show a wide impact
Rented, not returned	A user on a popular social app shared forgetting to return a bike: the card shows "rented" for about 4 days with an estimated fee near NT$9,000	User post (single event, unverified)	A bike rented out but not returned has no field in the station data
Bike locked but return failed (user experience)	The screen shows "return successful" then switches off while the app shows failure; customer service asks to re-tap the card for the charge or sends someone	Dcard and PTT discussions (user experience, not official)	Locked does not mean registered

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Another unseen kind: rented and never returned

Rental card (illustration)

Rented

Estimated accumulated fee

about NT$9,000

Subsidies excluded

Rental time about 4 days

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What this shows

A rented bike can stay away from stations for days: the card shows "rented" for about 4 days with an estimated fee near NT$9,000

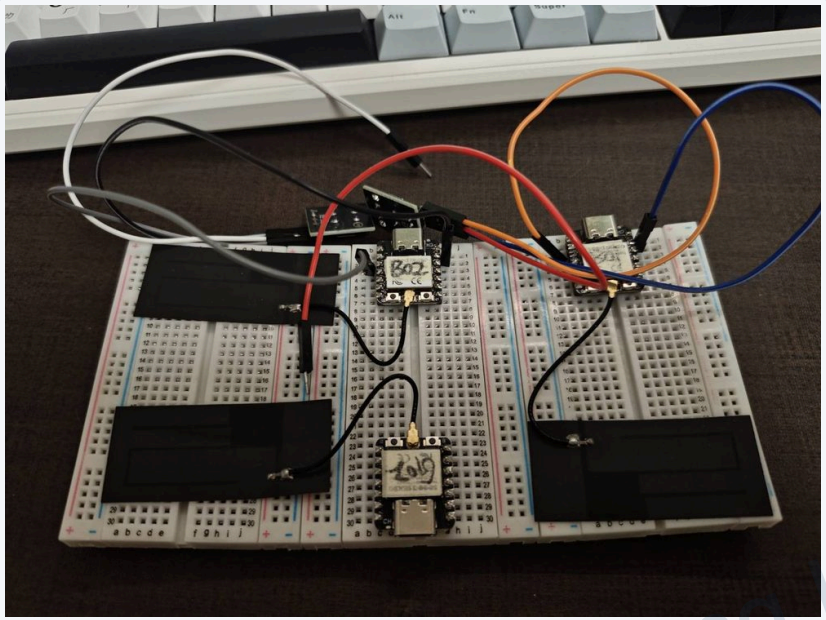
Such a bike is not on a dock, and the station data has no field to express it

A bike unit reports by itself, independent of docks. Concept: a temporary return is recorded in this study's own log at the moment it happens

A single event is not a distribution; it shows that bikes the docks cannot see have several origins

On a popular social app someone shared forgetting to return a bike (a single event); this is a redrawn illustration with no account or personal information.

Edge device: a virtual dock records a temporary return



Bench: three XIAO ESP32-C3 boards (external antennas), one labelled B02 (the bike unit), with a button module on top.

Bike unit (on the bike) — XIAO ESP32-C3 + button

Button press = temporary return (plus a periodic heartbeat)

↓

Wireless advertisement and pairing — (format not published)

Short Bluetooth packet with an authentication code

↓

Virtual dock (gateway near the station) — USB to computer

Rider's pairing point: de-duplication, cooldown, one temporary return

↓

Rider's temporary-return record — Task page / console

Aggregated into the off-dock estimate

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The bike unit sends its own signal and pairs with the virtual dock without relying on physical docks; it reports only: which station, how many, when, who. No unlocking, no lock control, no billing changes.

Verified so far (bench): bike unit → advertisement → virtual dock (gateway) (2026-08-28, short range); gateway → console (10-07 local event drill). Next: work with the operator on network upload, multiple boards and pairing with real bikes.

A temporary return is not an official return. Billing does not stop. Do not leave a bike outside the docks because of it. Reports are unverified signals and never drive dispatch on their own.

Edge reporting replaces the assumed off-dock count

Without edge reporting the off-dock count can only be a scenario assumption; the shortage estimate is highly sensitive to it, and that is the value of reporting.

Assumed off-dock bikes (citywide)	0	1,000	2,000	4,262*	6,000	8,000
Average shortage remaining, all slots (bikes)	314.9	129.5	46.5	4.75	0.93	0.16
Weekday 15:30	522.7	218.3	76.0	9.11	2.28	0.34
Weekday 10:00	146.9	62.1	29.0	3.72	0.76	0.28

Replay estimate over 280 slots (clean days 07-21 to 10-15, main 72 circles); the off-dock count is a scenario assumption; 4,262* is an assumed maximum.

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Information value of edge reporting

① Reports replace the assumption: the shortage estimate no longer rests on an assumed maximum

② Tell a real shortage from bikes outside the docks: expected (unverified) that motorbike staff can re-dock directly

③ Trucks do only what trucks should: direct moves within 2 km resolve only 8%—26% of the shortage

Indicators to compare after adoption

How much the shortage actually shrinks after reporting: compare with the operator's official data

Reporting error and miss rate

Stage reached: sensitivity analysis (algorithm layer) + link verification (device layer)

Cloud receiver (demo, removed 2026-10-21)



Task page (phone): station status / off-dock bikes / temporary-return point / broken bikes (the page UI is in Traditional Chinese).

One station shown (name blurred); the screenshot is demo data and the demo unit has been removed.

Architecture: the cloud is only a mailbox

Design: field reports go to a cloud mailbox, are pulled back to a local report file, and the console reads it

The demo unit has been removed; the screen and data were demonstration only, not a running service

The same task page also offers broken-bike reports and temporary-return points

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Peak and off-peak: each does its own job

Reportedly New Taipei requires balance within 2 hours after the peak; this study simulates moving the hauling to off-peak, so the peak only does things that need no vehicle movement.

Off-peak (four windows)

05:30, 10:00, 15:30 and 23:00 once each; 05:30 and 15:30 are completion deadlines

Few trucks, forward pre-positioning: inside 2 km circles, at most 14 bikes per truck, no backtracking

Reports accumulated at the peak (temporary returns, broken bikes) update the off-dock estimate; they never trigger dispatch on their own

Slack time: a looser deadline needs fewer trucks (chart on the right)

Peak

Peak-hour motorbike staff handle off-dock bikes (staff operations are outside this proposal)

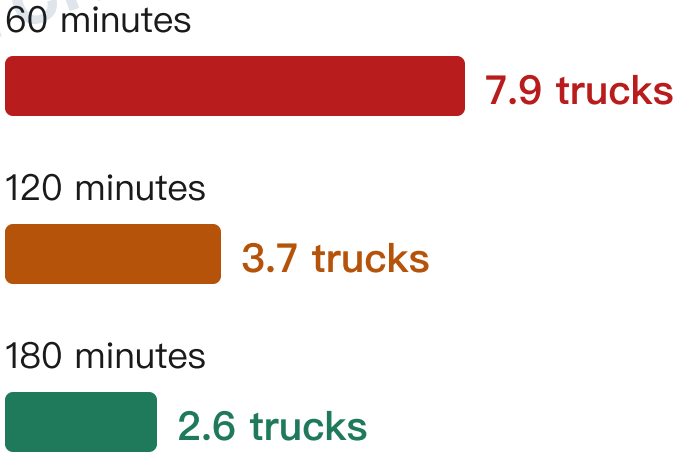
Re-docking off-dock bikes at the same time: only a layer change, totals unchanged, no vehicle movement

Trucks do not go out: peak congestion and high rental demand make it a poor time to haul

Pre-positioning off-peak puts the directly movable part in place before the peak; the rest depends on re-docking at peak

Same job: a longer deadline needs fewer trucks

Trucks per slot (96-circle replay; reserve assumption 4,262)



Assumptions: 40 km/h, detour 1.3, no stop time

Division of work: off-peak, few trucks over a long window; at peak, motorbike staff change the layer. A looser deadline means fewer trucks out at once, but total workload is about the same (about 7.4–7.9 truck-hours).

Numbers are replay estimates from official data; the deadline-to-trucks relation is a model output, not operating results.

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One button: output of one trial batch

Local console, one batch at 2026–10–21 18:59, not directly comparable with the 8%—26% replay average; window 23:00; algorithmic estimate

Main 72 circles: shortage 212.1 (15 circles) = 113.8 dispatched within 2 km + 98.3 from reserve bikes (scenario assumption); island groups: another 9.3 dispatched; reserve is an estimate

Top 6 circles receiving bikes	Bikes in
Fuzhong (MRT)	21
Huilong (MRT)	21
Far Eastern Hospital (MRT)	14
Haishan (MRT)	12
Yongan Market (MRT)	9
Ankang (LRT)	9

Top 6 flows: from (surplus circle)	To (short circle)	Bikes	Distance
Xinpu (MRT)	Fuzhong (MRT)	14	1.85 km
Banqiao Bus Station (TRA)	Far Eastern Hospital (MRT)	14	1.91 km
Danfeng (MRT)	Huilong (MRT)	13	1.34 km
Tucheng (MRT)	Haishan (MRT)	12	1.44 km
Zhonghe (MRT)	Yongan Market (MRT)	9	1.55 km
Xiaobitan (MRT)	Ankang (LRT)	9	1.5 km

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How many trucks, and when to leave

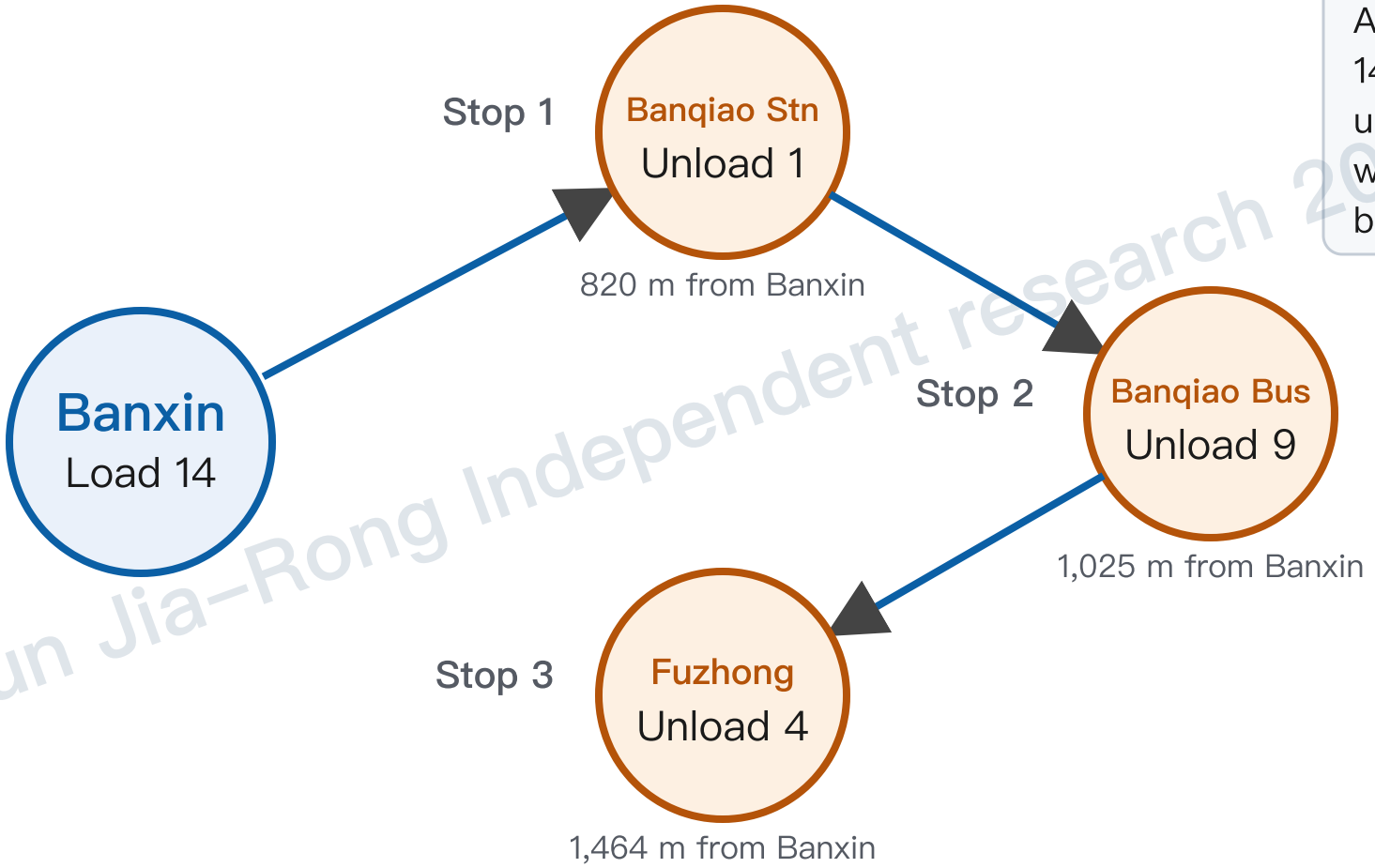
Within 120 minutes: optimistic 3, middle 4, conservative 4 trucks (Wanhua 2—3, Gongguan/Da'an 1); the 23:00 window needs the latest departure at 21:00 (trial)

Forward routes: 17 trips (incl. island groups), 123 bikes, 28.3 km; each leg at most 2 km, no backtracking, 14 per truck; a 180–minute deadline needs only 3 trucks

Recomputed automatically every 30 minutes; assumed 40 km/h, detour 1.3, no stop time

How one truck covers one trip: a trial route

A trial batch from 2026–10–08; schematic, positions not to scale; distances are straight lines between circle centres.



About 1.7 km in total, 14 bikes. All unloaded; every leg is within 2 km, no backtracking.

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Rules: at most 14 bikes per truck; each leg at most 2 km; no backtracking (one 120-degree sector).

Results: the largest shortage is the National Day afternoon

2026-07-21 to 10-15, 280 clean slots; the first two rows use the main 72 circles, the others all 96.

Item	Figure	Note
Stability on new days	Weekday remaining shortage: 0.6 to 3.6 bikes	Main 72 circles; weekday slots of the 10 days from 10-06 to 10-15, which the base table has not seen
Shortage resolved directly by trucks	8% to 26%	Main 72 circles; replay average; the rest is covered by reserve bikes (assumed)
National Day 15:30 (10-09 to 10-11)	Shortage 2,628; remaining 253 bikes (per-slot average of 3 days)	Left after ideal dispatch plus the assumed reserve; about 60% on island groups; 3 days; under the 4,262 scenario
Mid-Autumn 15:30 (09-25 to 09-28)	Shortage 493; remaining 9.8 bikes (per-slot average of 4 days)	Same method, no large shortage (4 Mid-Autumn days are already in the base table)
Trucks per slot	About 3.7	120-minute deadline; 7.9 at 60 minutes, 2.6 at 180 minutes

Replay report. Remaining shortage = not covered by ideal dispatch under the 4,262 reserve (fewer reserve bikes mean a larger shortage, more mean a smaller one). Weekday 0.6—3.6 comes from the weekday slots of the 10 out-of-sample days; estimate only.

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How the proposal maps to existing arrangements

Area	Current situation (public information)	How this proposal responds
Operating model	Operator and government have their own contracts; this proposal is not a party; the operator runs operations and dispatch	Makes no operator decisions; supports "how many, and how" decisions
Service-level requirement	New Taipei requires each district balanced within 2 hours after the peak; a fine of NT$1,000 per problem station (2025 reports: New Taipei press release 2025-09-01, United Daily News 2025-09-30)	Pre-position off-peak so bikes are in place before the peak; use 2 km circles to reduce post-peak recovery
Dispatch resources	New Taipei dispatch fleet 41 trucks, 350 staff (2025 reports, same sources); about 14 bikes per truck (operator's figure, United Daily News 2025-08-06)	Constraint of 14 bikes per truck; assumed 40 km/h, detour 1.3, no stop time; about 3.7 trucks per off-peak slot finishing within 120 minutes (96-circle replay)
Depots	Author's assumed departure points: three Taipei-side locations (Wanhua, Neihu, Gongguan/Da'an); coordinates are landmark approximations, not operator data	Assume trucks leave from these three points off-peak
Data	Official public live station data: bikes and free docks; off-dock bikes have no field	The bike unit fills in the "off-dock layer"; everything else uses public data

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Done (including demos and proof of concept)

Result	Time	Content
Official station data collected and replayed	From 2026–07–21	280 clean slots (to 10–15), 96 two–kilometre circles
Algorithm and automatic computation	2026–10	Recomputed automatically every 30 minutes; one button gives routes, truck count and latest departure; local console, task center and task page built
Field count	2026–09–17	One station: 25 docks, 20 bikes outside (photo has a timestamp)
Bike–unit bench verification	2026–08–28	Bike unit → advertisement → gateway (short range)
Event drill	2026–10–07	Gateway → console (local)
Cloud task page	Demonstrated 2026–10, removed 10–21	Field–report receiver (demo data); broken–bike reports and temporary–return points
Truck parameters	2025 public reports	New Taipei: 41 trucks, 350 staff, 14 bikes per truck

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Scope, future extensions and next steps

Scope

This is an algorithm-level solution: how many to move, how, and when to leave; actual dispatch is carried out by the operator

Within the existing arrangements: public official data, fleet size and capacity, off-peak pre-positioning

Off-dock bikes are first estimated with the maximum-reserve scenario and replaced by reports once edge reporting is live

Design principles: public data, edge reports carry no rider identity, no control of vehicles

Possible extensions

The same four fields (station, count, time, source) could connect to other mobile assets measured against fixed anchors, such as self-driving vehicles or drones; each needs its own regulation and authority approval.

Next steps (90-day verification path)

Oct—Nov: end-to-end verification of bike unit + gateway uploading over the network (a new receiver is needed)

Nov—Dec: with an operator, compare estimates against its official data in 2—3 two-kilometre circles

2027-01: use reports to correct reserve allocation; add stop time and the real road network

Collaboration (no one approached yet)

An operator, a city government or a research group to trial in two-kilometre circles

Work with research groups to train dispatch and data talent

Turn replay estimates into the field validation they need

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Appendix: data sources, privacy and compliance notes

Area	What this proposal does	To be confirmed
Data sources	Uses public government station data plus self-accumulated snapshots; other public information (such as operator notices) follows each source's terms	Licensing terms and citation formats of each source are being checked one by one; redistribution of snapshots and scraping of notices should be checked by counsel
Personal data	No rider identity or ride tracks collected; edge report fields are which station, how many, when, who reported (a reporter code)	Retention and deletion rules for the reporter code
Bikes and systems	No unlocking, no lock control, no billing changes; observe, report and estimate only	—
Report reliability	De-duplication and cooldown; reports expire	Measure error and missed reports when compared against operator data
Disclosure	Results are labelled "algorithmic estimate"; the off-dock count is a scenario assumption with a sensitivity table	The content published here contains no bike numbers

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